

Question Bank

DSA (MCA Sem-I)

Unit –I

1. Explain Time and space complexity.
2. What are the types of asymptotic notations used for algorithm analysis
3. Define data structure? Explain different types of data structures
4. Define Hashing? Explain collision resolution technique
5. Describe the concept of collision in a hash table and explain how collisions can be handled.
6. Explain difference between Big-O, Theta (Θ), and Omega (Ω) notations with examples of their use in algorithm analysis.
7. Explain the concept of hashing and describe any four hashing techniques
8. Difference between Linear and non-linear Data structure.
9. Explain Doubly linked list in detail.
10. Discuss different types of Collision Resolution Techniques.
11. What do you mean by Analysis of Algorithm.
12. Differentiate Best case and Worst case analysis in detail.

Unit- II

1. Explain Bubble Sort Algorithm with Example.
2. Write an algorithm for selection sort and Sort the following data :
6, 1, 4, 3, 5, 2, 7
3. Discuss the concept of Insertion sort with example.
4. Define Array as an Abstract Data Type (ADT).
5. Explain the binary search on a sorted array.
6. List the basic operations performed on array
7. Explain Linear search Algorithm in detail
8. Explain Merge sort Algorithm with example
9. Explain Binary Search algorithm in detail.
10. Sort the following of data using merge sort:
10, 9, 8, 7, 6, 5, 4, 3, 2, 1

Unit – III

1. What is circular queue ? Explain the overflow and underflow condition in array representation of circular queue
2. Define Priority Queue. Explain Array representation of priority Queue in Memory.
3. Write an algorithm for the postorder traversal of binary tree.
4. Describe an algorithm to insert a node at the beginning of a singly linked list
5. Write a short note on: —(i) Dequeue (ii) Priority Queue
6. Describe the algorithms used to perform push and pop operations in a stack.
7. Write an algorithm for removing element from queue which is represented as linked list
8. Explain the concept of Priority Queue with example.

9. Define Linked List with example
10. Differentiate between sequential and linked representations of lists.
11. Write an algorithm for removing element from queue which is represented as linked list
12. Write an algorithm for removing element from queue which is represented as linked list

Unit – IV

1. What is a Linked List? Explain the representation of single linked lists in memory
2. Write an algorithm for removing element from queue which is represented as linked list
3. What is Stack? What are the different operations that can be performed on stack ?
4. Describe the algorithm to insert a node at the beginning of a linked list.
5. Discuss the meaning of a queue and demonstrate the two ways it can be represented with examples.
6. Explain recursion with a suitable example and identify its base condition
7. Define a queue and explain the two methods of queue representation.
8. Analyze the postfix expression evaluation algorithm and explain its execution step-by-step using an example.
9. Evaluate the following postfix expression
 $2 \uparrow 3 + 5 * 2 \uparrow - 12 / 6$
10. Translate infix expression into its equivalent prefix expression
 (i) $(A-B) * (D/E)$.
 (ii) $(A+B \uparrow D)/(E-F)*G$.
11. What is recursion? How Tower of Hanoi problem can be solved using recursion ?
12. Translate the following infix expression in prefix and postfix notation :
 (i) $a + (b - c) * d / (e * f)$
 (ii) $(a + b * c) / ((a + b) * c)$
13. Write an algorithm to evaluate the postfix expression and explain with example
14. Explain Tower of Hanoi problem with example.
15. Explain PUSH and POP operations with example.

Unit- V

1. Describe the concept of a tree and illustrate how a binary search tree organizes its nodes.
2. Describe the concept of BFS, DFS.
3. Explain BFS method of traversing graph with suitable example.
4. Construct examples of a graph in both linked-list and matrix forms, and generate a report comparing their performance for different graph types
5. Explain the applications of graphs in real-world problems like networking and routing.
6. Illustrate the working of Kruskal's algorithm for finding a minimum spanning tree with a suitable example
7. What is a minimal spanning tree? Mention one application.
8. Explain Prim's Algorithm in detail.